Complex Tori and Elliptic Curves

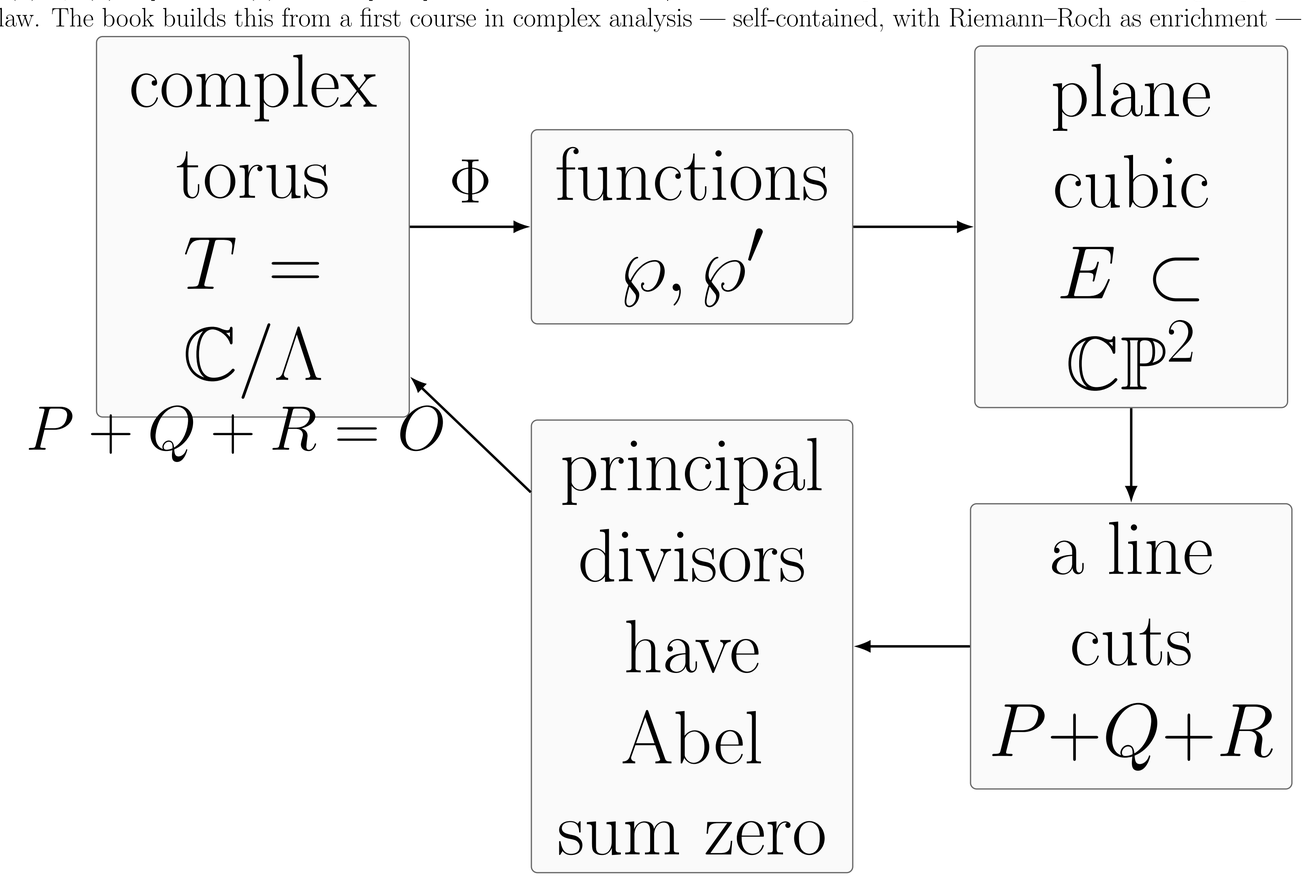
A complete synopsis — every Part, chapter, and result of the textbook

The chord–tangent law on a smooth cubic is *addition on* $\mathbb{C}/\Lambda$, *transported by* $\wp, \wp'$

The Main Theorem and Its Logic (Part I, Ch. 1)

addition on $\mathbb{C}/\Lambda \xrightarrow{\sim}$ chord–tangent law on E $\qquad E:\ Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3 \subset \mathbb{CP}^2$

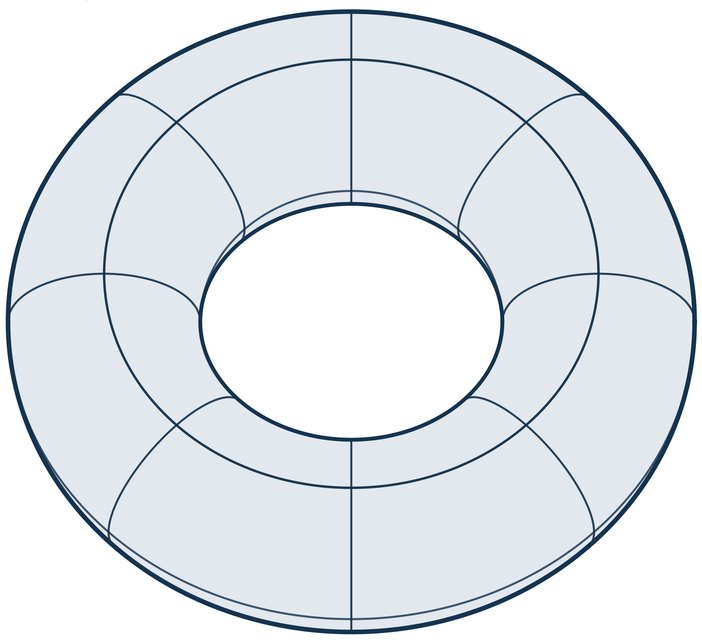
Theorem. The Weierstrass map $\Phi([z]) = [\wp(z) : \wp'(z) : 1]$, with $\Phi(0) = O = [0{:}1{:}0]$, is an isomorphism $\mathbb{C}/\Lambda \xrightarrow{\sim} E$ of compact Riemann surfaces *and* of groups: ordinary addition modulo a lattice becomes the chord–tangent law. The book builds this from a first course in complex analysis — self-contained, with Riemann–Roch as enrichment — along the chain:



Part I | Tori as Riemann Surfaces

Ch. 2 Lattices. A lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ has period ratio $\tau = \omega_2/\omega_1 \in \mathbb{H}$, on which $\mathrm{SL}_2(\mathbb{Z})$ acts by Möbius maps; homothety classes form the moduli space $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. The charts are local inverses of $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ on disks of radius $< \frac{1}{2}m(\Lambda)$, with translation transition maps: $T = \mathbb{C}/\Lambda$ is a **compact Riemann surface**.

Ch. 3 Group & surface. Addition $[u] + [v] = [u + v]$ and negation are holomorphic, so T is a complex Lie group on which translations act simply transitively. Topologically $T \cong S^1 \times S^1$; a CW count gives $\chi(T) = 0$, hence **genus** $g = 1$ — the topological fact behind the simple form of Riemann–Roch.



Part II | Elliptic Functions

Ch. 4 Liouville. Elliptic functions are doubly periodic meromorphic functions; a nonconstant one has poles (order ≥ 2) and takes each value N times. Integrating f, f'/f, $z\,f'/f$ around a period parallelogram (opposite sides cancel), $\oint f'/f\,dz = 0$ gives #zeros= #poles, and the non-periodic z leaves $\oint z\frac{f'}{f}\,dz = 2\pi i(k_2\omega_1 - k_1\omega_2)$, the **Abel relation**

$$\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda.$$

Ch. 5 Divisors. $\deg \mathrm{div}(f) = 0$ on a compact surface (residue theorem on df/f). The Abel–Jacobi sum map $\mathfrak{s} : \mathrm{Div}^0(T) \to T$ kills principal divisors and descends to an isomorphism $\mathrm{Pic}^0(T) \cong T$ — the torus is its own Jacobian.

Part III | Riemann–Roch

Ch. 6. $L(D) = H^0(\mathcal{O}(D))$, $\ell(D) = h^0$. The counts $\ell(O) = \ell([a]) = 1$ and $\ell(nO) = n$ are *elementary* (Liouville + the explicit $\wp, \wp'$), forcing the functions $1, \wp, \wp'$. With trivial canonical $K \sim 0$, Riemann–Roch reads $\ell(D) - \ell(-D) = \deg D$; the σ-converse of Abel completes $\mathrm{Pic}^0(T) \cong T$. Riemann–Roch is kept as the general theorem, but is needed nowhere.

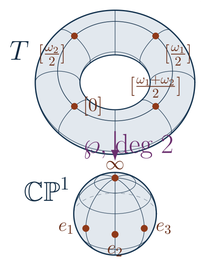
Part IV | The Weierstrass Functions

Ch. 7 $\wp$. $\wp = \frac{1}{z^2} + \sum_{\lambda \neq 0}\left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2}\right)$, even of order 2; $\wp'$ odd of order 3; Laurent $\wp = z^{-2} + 3G_4z^2 + 5G_6z^4 + \cdots$. Squaring and cubing, the poles cancel:

$$\wp'^2 - 4\wp^3 = -60G_4z^{-2} - 140G_6 + O(z^2),$$

so with $g_2 = 60G_4$, $g_3 = 140G_6$, $\boxed{\wp'^2 = 4\wp^3 - g_2\wp - g_3}$. Roots $e_i = \wp(\omega_i/2)$ are distinct, $\Delta = g_2^3 - 27g_3^2 \neq 0$; the σ-function gives the **constructive converse of Abel**.

Ch. 8 Embedding. The complete system $|3O|$ $(1, \wp, \wp')$ gives the closed immersion $\Phi = [\wp : \wp' : 1]$, $\Phi(O) = [0{:}1{:}0]$, a **biholomorphism** $T \xrightarrow{\sim} E$; the inverse $z \mapsto -z$ is reflection $(x, y) \mapsto (x, -y)$.



The map $\wp : T \to \mathbb{CP}^1$ is a 2 : 1 branched cover with 4 branch values: three roots e_1, e_2, e_3 and ∞. Riemann–Hurwitz: $0 = 2{\cdot}2 - 4{\cdot}(2-1)$.

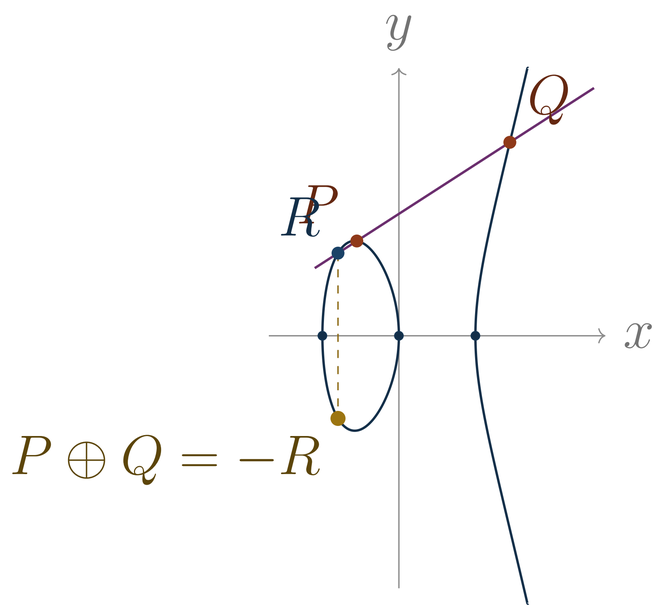
Part V | The Group Law on the Cubic

Ch. 9 Abel's law. A divisor is principal $\iff$ $\deg = 0$ *and* Abel sum $= 0$; in three-point form, $[p] + [q] + [r] - 3[0]$ is principal $\iff$ $p + q + r \equiv 0$.

Ch. 10 Lines. A line restricts to ℓ/Z with $\mathrm{div}(\ell/Z) = P{+}Q{+}R{-}3O$ (Bézout, the flex at O): P, Q, R **collinear** $\iff$ $P \oplus Q \oplus R = O$.

Ch. 11 Transport. The transported torus law *equals* the chord–tangent law (forced by three choiceless facts), and is intrinsic to (E, O) — associativity inherited from $\mathbb{C}$.

Ch. 12 Cases. Identity $P \oplus O = P$; chord $P \oplus Q = -R$; tangent $2P = -R$; vertical lines give inverses; two-torsion $E[2] = \{O, e_1, e_2, e_3\} \cong (\mathbb{Z}/2\mathbb{Z})^2$.



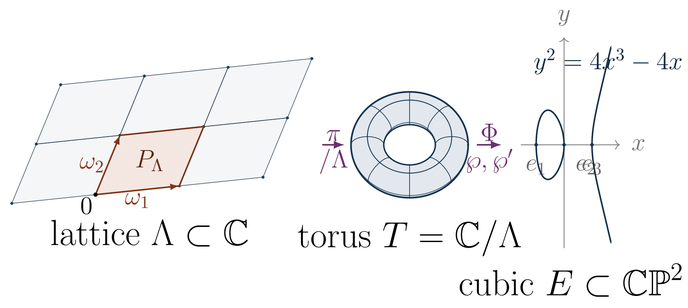
Part VI | Formulas, Computations & Synthesis

Ch. 13 Coordinates. Substituting $y = mx + b$ into the cubic and applying Vieta, $x_{P\oplus Q} = \frac{m^2}{4} - x_1 - x_2$, $y = -(mx + b)$, with $m = \frac{y_2 - y_1}{x_2 - x_1}$ (chord) or $\frac{12x_1^2 - g_2}{2y_1}$ (tangent). The law is **rational**, so $E(k)$ is a group over *any* field — the gateway to arithmetic.

Ch. 14 Analytic addition. $\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}\right)^2$, proved from Abel and the ODE; the reflected y-sign is the oddness of $\wp'$; $v \to u$ gives the duplication formula.

Ch. 15 Worked & pitfalls. Exact run on $y^2 = 4x^3 - 4x$: $P \oplus Q = (-\frac{24}{25}, -\frac{28\sqrt{6}}{125})$, heights grow fast ($3P = (\frac{2738}{529}, \cdots)$). The rescale $y = 2Y$ gives $y^2 = x^3 + ax + b$; six recurring pitfalls.

Ch. 16 Synthesis. The exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_E \to \mathcal{O}_E^* \to 0$ gives $\mathrm{Pic}^0(E) = H^1(\mathcal{O}_E)/H^1(\mathbb{Z}) \cong \mathbb{C}/\Lambda$: **one abelian variety, three incarnations** $\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(k)$. Then $j = 1728\,g_2^3/\Delta$ and moduli $\mathcal{M}_{1,1}$, isogenies $E[n] \cong (\mathbb{Z}/n)^2$, $E(\mathbb{Q})$ and $E(\mathbb{F}_p)$, modular forms and modularity.



Appendices & Apparatus

A. Complex-analysis toolkit: Liouville, residues, the residue theorem, the argument principle — the single course the book assumes.

B. Exam-to-lecture bridge: how the material grew from a course's midterm and final.

C. The Sage→TikZ pipeline: every figure here is drawn from coordinates computed by `sagemath/`, fully reproducible.

D. About 80 exercises, all with complete worked solutions.

Author: ⟨Your Name⟩ • ⟨email⟩ • seeking PhD positions in arithmetic and algebraic geometry. Textbook *Complex Tori and Elliptic Curves* (∼130 pp, self-contained, reproducible Sage figures).

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